

## Highlights

### **Forecasting the impact of the viral ‘Kia Boys’ social media trend on car thefts to 2050**

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- Social media is widely seen as a potent catalyst of harmful offline behavior.
- A viral online trend linked to a surge in thefts of Hyundai and Kia cars across the United States in 2022 played a muted role in the surge in Los Angeles, which began in mid-2020.
- By characterizing the offline dynamics using cars popular among previous generations of car thieves we project elevated thefts of vulnerable cars through at least 2042.
- Higher-income areas of Los Angeles will have long forgotten about the ‘Kia Boys’ vulnerability while lower-income areas will continue to absorb vulnerable model-year cars as they move towards their end of life-cycle.

# Forecasting the impact of the viral ‘Kia Boys’ social media trend on car thefts to 2050

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## ABSTRACT

In 2022, Hyundai Sonatas became the most frequently stolen model of car in Los Angeles, mirroring a nation-wide trend. The sudden popularity of this mid-sized sedan among car thieves has been linked by some to the viral spread of videos on TikTok and YouTube that showcased a security vulnerability making certain Kia and Hyundai models easy to steal. These videos later received the moniker ‘Kia Boys’ or ‘Kia Challenge.’ In this work, we show that the surge in Hyundai Sonata thefts in Los Angeles was only weakly tied to the ‘Kia Boys’ trend, arising much earlier during COVID-19 pandemic lockdowns. Any lasting effects of the viral social media trend were fully absorbed into offline car theft dynamics by the time social media interest peaked in 2022. We use historical data on Honda Civics and Toyota Camrys, car models popular among pre-social media generations of car thieves, to develop a model for the offline temporal and spatial evolution of theft preferences. The model suggests that the effects of the ‘Kia Boys’ vulnerability are likely to persist for decades irrespective of the role social media played in its inception and that these effects will be increasingly concentrated in low-income communities over time.

## 1. Introduction

Car theft remains a significant challenge for most US metropolitan areas, though it rarely generates news headlines (Tapp and Coen, 2024; Farrell, 2025). For example, across 9,639 articles in *The New York Times* in 2019 containing the keyword “crime” less than 0.2% ( $n = 18$ ) mention “car/auto/vehicle theft” or “stolen car/vehicle.” The recent surge in thefts of Kia and Hyundai car models is a stark exception (Endale, 2022; Casey and Dohr, 2022; Morrison, 2023). Widespread media reporting, starting in June 2022, raised an alarm about videos posted on TikTok and YouTube under labels such as the ‘Kia Boys’ and the ‘Kia Challenge’ that showed how to exploit a unique mechanical vulnerability that made several Kia and Hyundai car models easy to steal with little more than a USB key. In this study, we examine the recent surge in thefts of Hyundai Sonatas in Los Angeles, CA, to become the top-ranked stolen car, and use a comparative model, based on previous generations of top-ranked stolen cars, to forecast the temporal and spatial impact of the ‘Kia Boys’ vulnerability through 2050. The results suggest not only that the role of online social media in generating this offline harm is more nuanced than popularly portrayed but also that ignoring key offline mechanisms to focus on unique online processes would lead to a substantial mischaracterization of the dynamical system over the long-term. Model-based forecasts suggest that thefts of the vulnerable Hyundai and Kia models in Los Angeles will remain elevated over baseline levels through at least 2042,

dropping to negligible levels between 2043-2050 largely because vulnerable vehicles will have become rare due to scrapping. The long-term impact will vary across the city, however. Higher-income areas of Los Angeles will have forgotten about the ‘Kia Boys’ vulnerability as older, at risk cars are traded in for newer ones. Lower-income areas of the city will continue to absorb vulnerable cars as they change hands on the used market before moving towards eventual scrapping.

The remainder of this paper is organized as follows. In Section 2 we provide historical background on vehicle anti-theft technologies and the origin of the ‘Kia Boys’ social media meme or trend. Section 3 introduces our data and the mathematical model used for long-term theft forecasting. Section 4 turns to results, focusing on the trajectory followed by the Hyundai Sonata to become the top-ranked stolen car in Los Angeles and then turning to historical analogs for popular targets among Los Angeles’ car thieves. We conclude with a discussion of the role of online and offline dynamics in the rise of the ‘Kia Boys’ phenomenon.

## 2. Background

### 2.1. Vehicle Anti-theft Technology

Many modern cars include a range of anti-theft technologies that are meant to thwart common methods car thieves use to gain entry to and start a vehicle without the correct key (Farrell, 2025). In addition to ubiquitous audible alarms, many cars now include keyless entry systems, locking steering wheels, chip-based engine immobilizers or pushbutton ignition systems. Part marking and GPS-based vehicle tracking systems are additional technologies used to deter offenders and to aid in investigation and recovery if a vehicle

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has been stolen. Engine immobilizers have been the most successful of these several anti-theft technologies (Sallybanks and Brown, 1999; Brown, 2015; Kriven and Ziersch, 2007). Engine immobilizers were first widely deployed in the U.S. car market in the 1990s following Congressional approval of the 1984 Motor Vehicle Theft Law Enforcement Act (MVTLEA) (42 U.S.C. §14171) and its expansion under the 1992 Anti-Car Theft Act (ACTA) (H.R. 4542, revised in 1996) (see Farrell, 2025). Most modern engine immobilizers use a challenge-response protocol that compares a code generated by a transponder embedded in the key to the code expected by the ignition system (Davis and DeLong, 1996). In early systems, each key associated with a vehicle issued a unique encrypted RF signal that was decoded by the ignition system and compared with authorized keys. More recent immobilizers generate a new encrypted “rolling code” each time the key is used (Davis and DeLong, 1996; Brown, 2017), much like website multi-factor authentication systems. Absent a correctly paired key, the engine will not start. Of course, no practical security system can guarantee protection against theft (Wouters, Van den Herrewegen, D Garcia, Oswald, Gierlichs and Preneel, 2020).

Unlike in the UK and Australia (Brown, 2004; Farrell, 2025), the MVTLEA and ACTA did not require car manufacturers to include immobilizers as standard equipment in U.S.-sold cars. Manufacturers were instead required to stamp or mark critical car parts, with the goal of making those parts less desirable to so-called “chop shops” (see Harris and Clarke, 1991). Manufacturers could receive an exemption from the parts marking requirements for specific car models if they could demonstrate that thefts of the model were likely to fall below the median theft rate for passenger vehicles (3.2712 thefts per 1,000 vehicles based on 1985 data, under MVTLEA, 3.5826 thefts per 1,000 vehicles based on 1990-1991 data, under ACTA). New models of high theft lines could also be exempted if the new models included proven anti-theft technologies such as an immobilizer as standard equipment (Department of Transportation, 1999). Thus, incentives in the MVTLEA and ACTA were structured to encourage the adoption of immobilizers or other effective theft prevention technologies in place of costly parts marking. As discussed in further detail below, Toyota introduced immobilizers on many vehicle models beginning in 1992 and Honda did so in 2001. In spite of the incentives, however, not all vehicles with immobilizers were immediately granted exemption from part marking requirements.

Hyundai and Kia entered the U.S. car market in 1986 and 1997, respectively (Jacobs, 2023), and engine immobilizers were introduced on several models across the fleet in the mid-2000’s. For example, the repair manual for the 2004 model-year Hyundai Sonata identifies a chip-based immobilizer as standard equipment on the entry-level model. Beginning in 2011, however, 11 Hyundai and nine Kia models were sold in the U.S. without engine immobilizers or with their functionality disabled (Kia Hyundai Vehicle Theft Litigation, 2023a). Absent a functioning immobilizer, these cars proved to be easy to steal in a matter of minutes.

| Hyundai         |              | Kia      |       |
|-----------------|--------------|----------|-------|
| Model           | Years        | Model    | Years |
| Accent          | 11-22        | Forte    | 11-21 |
| Elantra         | 11-22        | K5       | 21-22 |
| Elantra GT      | 13-20        | Optima   | 11-20 |
| Elantra Coupe   | 13-14        | Rio      | 11-21 |
| Elantra Touring | 11-12        | Sedona   | 11-21 |
| Genesis Coupe   | 11-14        | Seltos   | 21-22 |
| Kona            | 18-22        | Soul     | 11-22 |
| Palisade        | 20-21        | Sorento  | 11-22 |
| Santa Fe        | 11-22        | Sportage | 11-22 |
| Santa Fe Sport  | 13-18        |          |       |
| Santa Fe XL     | 19           |          |       |
| Sonata          | 11-19        |          |       |
| Tucson          | 11-22        |          |       |
| Veloster        | 12-17, 19-21 |          |       |
| Venue           | 20-21        |          |       |
| Veracruz        | 11-12        |          |       |

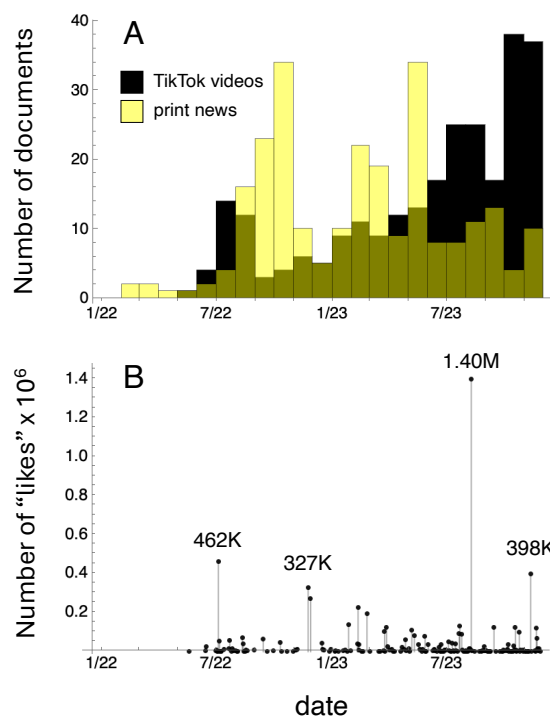
**Table 1**

Vulnerable Hyundai and Kia models and model-years identified in the class action settlement *Kia Hyundai Vehicle Theft Litigation* (2023b). Model-years listed in two-digit form (e.g., 11-22 is 2011-2022).

Theft of one of these vulnerable cars proceeds through a sequence of steps including gaining entry without triggering the alarm, opening the steering column casing, disassembling or breaking the ignition cylinder to expose the ignition switch, and mechanically turning the ignition switch to start the car (Kia Hyundai Vehicle Theft Litigation, 2023a; Kriven and Ziersch, 2007). In the vulnerable Hyundai and Kia models, the exposed ignition switch is a small metal post that serendipitously matches the female-end of a USB-B cable, a ubiquitous item that makes for an ideal opportunistic tool. The vulnerable car models were updated by Hyundai and Kia to eliminate the vulnerability by the 2022 model-year at the latest. Some owners of vulnerable cars became eligible to receive free software updates or physical security devices (e.g., steering wheel locks) to address the problem in February 2023, as part of a class action lawsuit filed in 2022 (Kia Hyundai Vehicle Theft Litigation, 2023b; Highway Loss Data Institute, 2024). Several details of the class action suit remain under appeal and the uptake of agreed-upon security fixes remains uncertain. An additional multi-state settlement was announced in December 2025 requiring Hyundai and Kia to replace ignition cylinders in vulnerable vehicles (Compton, 2025). The expected implementation time for the multi-state settlement stretches into 2027.

## 2.2. Discovery of the ‘Kia Boys’ Vulnerability

While several Hyundais and Kias manufactured and sold after 2011 were uniquely at high risk of being stolen, what is now labeled the ‘Kia Boys’ vulnerability appears not to have been recognized by car thieves for at least a decade. Local media reporting in Milwaukee, WI, suggests that the vulnerability was first discovered sometime in mid-2020 (Dortch, 2020; ABC News, 2021). The earliest media reports do not

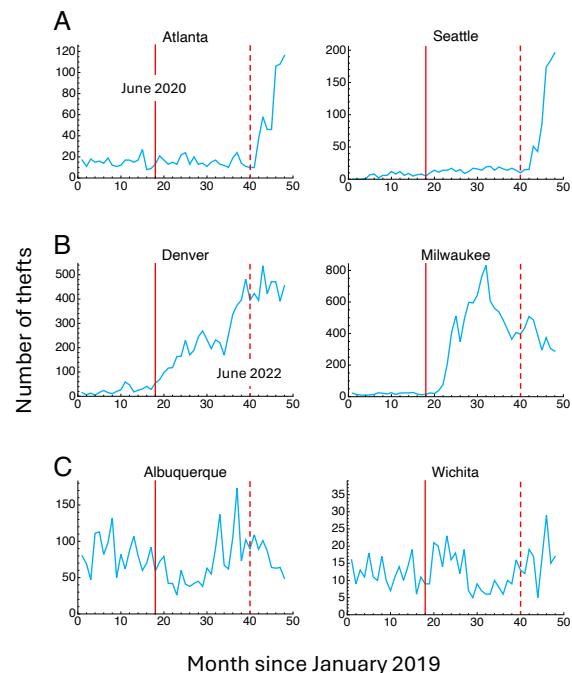


**Figure 1:** ‘Kia Boys’ in U.S. print media and on TikTok and. A. Number of TikTok videos and print news stories mentioning the ‘Kia Boys.’ B. Number of “likes” of the TikTok videos shown in A.

attribute the surge in thefts to viral online content, however. The oldest video still posted on YouTube referencing the ‘Kia Boys’ dates to June 5, 2021 (YBKQUIN, 2021). Viral spread of the meme online was likely initiated much later, in June 2022, through national news coverage by mainstream media (Fig. 1A,B, and Fig. 3A).

Thefts of vulnerable Kia and Hyundai models surged after June 2022 in a large number of cities including Atlanta, GA, and Seattle, WA (Fig. 2A) (USAFacts, 2023), a point also observed in insurance-claims data for several states (Highway Loss Data Institute, 2024). Car thieves in these places were apparently unaware of the vulnerability until it went viral in 2022. In other cities including Milwaukee, WI, and Denver, CO, the vulnerability appears to have been discovered independently during the summer of 2020, without the influence of a viral social media trend (Fig. 2B) (USAFacts, 2023; Highway Loss Data Institute, 2024). As discussed in detail below, Los Angeles, CA, appears to fall into this “early discovery” group (compare Fig. 2B, middle row, with Fig. 3A). In a handful of cities, there was no obvious change in thefts through the end of 2023 (Fig. 2C).

In our study, we examine the growth in thefts of Hyundai Sonatas in Los Angeles, CA, in relation to the rise of the ‘Kia Boys’ social media phenomenon and then use a comparative mathematical model to forecast long-term temporal and spatial pattern of thefts through 2050.



**Figure 2:** Monthly number of thefts of combined Hyundai and Kia models in six U.S. cities. A, Top Row. Cities with late onset of thefts. B, Middle Row. Cities with early onset of thefts. C, Bottom Row. Cities with no apparent change. Solid red lines mark June 2020. Dashed red lines mark June 2022. Aggregate theft data from 23 U.S. cities available from USA Facts.

### 3. Methods

#### 3.1. Data

Data on cars reported stolen in the City of Los Angeles were provided by the Los Angeles Police Department (LAPD). Between 2000 and 2023, a 24-year time period, 565,333 vehicles of all types were reported stolen (Table 2). The vast majority of these thefts are of passenger cars (e.g., Honda Civic, Toyota Camry), crossovers (e.g., Ford Escape, Mazda CX7), SUVs (e.g., Cadillac Escalade, Ford Bronco), and mass market light pickup trucks (e.g., Ford F150, Dodge Ram). For example, the top ten car makes by theft volume—Chevrolet, Honda, KIA, Toyota, Ford, Hyundai, GMC, Nissan, BMW, and Mercedes—alone account for 78.8% ( $n = 19,952$ ) of the 25,307 thefts reported in 2022. These annual totals likely underestimate the total number of thefts by 10 – 20% due to crime underreporting (Xie, Ortiz Solis and Chauhan, 2024). Nevertheless, the totals illustrate that car theft in Los Angeles declined substantially between 2000 and 2013 before surging again during two periods in 2014–2017 and 2019–2023.

We focus on vehicle model-year for specific car makes in our analyses (e.g., 2017 Hyundai Sonata, 1999 Honda Civic). Vehicle model-year is not only the most consistent level data recorded by police for reported thefts (i.e., trim level is haphazardly recorded), it is also the taxonomic scale at which offenders forage for targets and changes in car design are most likely to impact ease of theft and desirability



**Table 2**

Number and rank-order position of Hyundai Sonatas, Honda Civics, and Toyota Camrys stolen in Los Angeles by observation year.

| Year  | Hyundai Sonata | Honda Civic | Toyota Camry | All Vehicle Thefts |
|-------|----------------|-------------|--------------|--------------------|
| 2000  | 22 (143)       | 1,343 (5)   | 2,399 (2)    | 28,935             |
| 2001  | 30 (145)       | 1,969 (3)   | 3,109 (1)    | 31,825             |
| 2002  | 16 (234)       | 2,271 (4)   | 3,830 (1)    | 34,522             |
| 2003  | 30 (148)       | 2,161 (4)   | 4,151 (1)    | 33,543             |
| 2004  | 28 (151)       | 2,004 (3)   | 3,633 (1)    | 30,104             |
| 2005  | 22 (175)       | 1,969 (3)   | 3,347 (1)    | 28,737             |
| 2006  | 18 (178)       | 2,210 (3)   | 2,928 (1)    | 26,532             |
| 2007  | 21 (168)       | 1,970 (3)   | 2,493 (1)    | 25,189             |
| 2008  | 30 (121)       | 2,367 (2)   | 2,156 (3)    | 24,372             |
| 2009  | 18 (154)       | 2,289 (2)   | 1,363 (3)    | 19,547             |
| 2010  | 20 (128)       | 2,000 (2)   | 1,309 (3)    | 18,213             |
| 2011  | 19 (131)       | 1,845 (2)   | 1,008 (3)    | 16,790             |
| 2012  | 24 (98)        | 1,967 (2)   | 961 (3)      | 16,801             |
| 2013  | 36 (71)        | 1,974 (2)   | 704 (3)      | 15,925             |
| 2014  | 55 (46)        | 1,977 (2)   | 807 (3)      | 15,929             |
| 2015  | 63 (48)        | 2,573 (2)   | 786 (3)      | 18,833             |
| 2016  | 113 (28)       | 2,556 (1)   | 742 (3)      | 21,130             |
| 2017  | 111 (35)       | 2,362 (1)   | 665 (3)      | 21,693             |
| 2018  | 132 (22)       | 1,776 (1)   | 553 (4)      | 19,475             |
| 2019  | 115 (29)       | 1,342 (1)   | 500 (4)      | 17,434             |
| 2020  | 286 (14)       | 1,692 (1)   | 527 (6)      | 22,592             |
| 2021  | 943 (4)        | 1,377 (2)   | 573 (6)      | 25,411             |
| 2022  | 1,506 (1)      | 1,089 (4)   | 485 (8)      | 25,307             |
| 2023  | 1,642 (1)      | 932 (4)     | 479 (11)     | 26,494             |
| Total | 5,300          | 46,015      | 39,508       | 565,333            |

(Brantingham, 2013; Light, Nee and Ingham, 1993; Clarke and Harris, 1992). We selected from reported thefts the make, model, and model-year of all vehicles identified as Hyundai Sonatas, Honda Civics and Toyota Camrys. As shown below, Honda Civics and Toyota Camrys provide historical analogs for uniquely vulnerable cars that can be tracked over decades in theft data. Specifically, over the 24-year period of observation, 39,508 Toyota Camrys and 46,015 Honda Civics were reported stolen in the city of Los Angeles (Table 2). Over the same period, 5,300 Hyundai Sonatas were stolen. The rank-order position for each focal make-model, in each year of observation, was determined relative to thefts all unique make-models stolen during that year (Table 2). Thus, Honda Civics were in among the top-five most stolen make-models in all 24 years and the most popular target (i.e., Rank 1) of Los Angeles car thieves in five consecutive years (2016-2020). Toyota Camrys were among the top-five most stolen make-models in 20 of the 24 years and the most popular target in seven consecutive years (2001-2007). Hyundai Sonatas were among the top-five most stolen make-models during three years and were the most popular target in 2022 and 2023, the last two years of observation.

### 3.2. Mathematical Model

Our mechanistic, long-term forecasting approach is similar to compartmental model forecasts in epidemiology

(Brauer, 2008). Here, the macroscopic, qualitative long-term behavior of the system is the main interest. The forecasting model takes into consideration the entry of new model-year cars during each year of observation, the loss of old model-year cars through so-called ‘scrapping,’ and the timing of the introduction (or discontinuation) of specific anti-theft technologies such as immobilizers. When fit to observational data, the model provides a very general mechanistic framework that can serve as a baseline for forecasting vehicle theft trends more broadly.

We model theft patterns independently for each unique car make-model type and focus on trends in thefts by car model-year  $m$  (e.g., 2019 Hyundai Sonata is different from the 2020 Sonata). We first assume that the rate of car theft in year  $t$  is proportional to the abundance of model-year  $m$  vehicles in the environment (Appendix B):

$$\lambda_m(t) = h_m f_m(t) g_m(t) \quad (1)$$

where  $h_m$  is the number of model-year  $m$  cars sold in year  $t = m$ . We assume the number of cars sold increases as a power-law over time (since 1975)  $h_m = (t_m - 1975)^d$ , where 1975 is chosen as an arbitrary starting year and  $d$  scales the rate of market expansion. In Appendix C, we present model fitting results for several alternative specifications for  $h_m$  including constant, exponential, and Gaussian functional

forms as well as a data-driven model based on empirical sales data for each year  $m$ .

The abundance of car model-year on the street decays due to the scrapping of older cars, which we model as a logistic function of the time since the model-year was introduced  $t - t_m$  (Bento, Roth and Zuo, 2018; Greene and Chen, 1981):

$$g_m(t) = [1 + ae^{b(t-t_m)^c}]^{-1} \quad (2)$$

Equation (2) can be thought of as a survival function with a logistic functional form. Parameters  $a$ ,  $b$  and  $c$  determine the shape of the function and are fit from data. Generally, since  $b > 0$ ,  $g_m(t) \rightarrow 0$  as  $t - t_m \rightarrow \infty$ , meaning that cars of model-year  $m$  become less likely to survive or, equivalently, more likely to be scrapped as they age. The combined theft rate for all model-years of a car in year  $t$  is the sum of the rate for the individual model-years  $\lambda(t) = \sum_{m \leq t} \lambda_m(t)$ .

Next we model changes in theft vulnerability as a step function. We define  $s$  as the year in which a security vulnerability is introduced or removed from a car model. Then, at times  $t \geq s$  the constant  $f_m$  in Equation (1) increases or decreases by a certain amount:

$$f_m(t) = (f_0 + f_1 \chi\{t \geq s\}) \quad (3)$$

where  $\chi$  is an indicator function. If  $f_m(t) > f_0$ , a vulnerability was introduced in year  $t = s$  that increased the rate of thefts relative to some baseline  $f_0$ , whereas if  $f_m(t) < f_0$  then a defensive measure was introduced that decreased thefts. For Honda Civics and Toyota Camrys,  $s$  corresponds to the model-years  $s = 2001$  and  $s = 1992$ , respectively, when ignition immobilizers were added as standard equipment. All model-years prior to  $s$  lacked this defense, all model-years starting with  $s$  had the defense.

Hyundai Sonatas are more complex because there was a date at which a particularly vulnerable ignition system was introduced  $s_1 = 2011$  and a date at which it was eliminated  $s_2 = 2020$  (see Table 1). In addition, the vulnerability apparently was not discovered by car thieves in Los Angeles until mid-2020 (see below). To model the dynamics of the theft of Sonatas, we therefore modify Equation (3) as follows:

$$f_m(t) = f_0 + f_1 \chi\{t_m \geq s_1\} \chi\{t_m \leq s_2\} \chi\{t \geq u\} \sigma(r_0(t - u - x_0)) \quad (4)$$

Here, the increase  $f_0 + f_1$  applies to vehicles manufactured between  $s_1 = 2011$  and  $s_2 = 2019$ . However, the vulnerability only comes into effect in year  $u = 2020$ . The term  $\sigma(r_0(t - u - x_0))$  models logistic growth in the knowledge of the vulnerability among car thieves, which we argue began through exploration at the beginning of the COVID-19 pandemic, but may have been primarily influenced by viral videos in other places.

We also forecast the intensity of Hyundai Sonata thefts in different spatial regions of Los Angeles. The term  $f_m^b(t)$  partitions the intensity in region  $b$ , year  $t$  and car model-year  $m$  by assuming a separable model of the form:

$$f_m^b(t) = f_m(t) \cdot p_{t,m}^b \quad (5)$$

The theft fraction parameter  $p_{t,m}^b$  is fit from historical data for other top-ranked stolen cars. For example, we assume that the empirical fraction  $p$  of Honda Civics of model-year  $m$  stolen in LAPD's South Bureau is a function only of the age of the car,  $t - t_m$  (i.e., time since the model-year was introduced):

$$p_{t,m}^s = q(t - t_m) \quad (6)$$

and that the fraction of thefts in LAPD's West Bureau is a time-shifted version of  $q$ :

$$p_{t,m}^w = q(t - t_m + \tau). \quad (7)$$

We then use local polynomial regression to estimate  $q$ , where the offset used is  $\tau = 15$ .

We use STAN to fit Equations (1) and (4) (Carpenter, Gelman, Hoffman, Lee, Goodrich, Betancourt, Brubaker, Guo, Li and Riddell, 2017). STAN is a probabilistic programming language that allows for Bayesian inference of statistical model parameters. Because all parameters in the models are positive, we used weakly informative exponential priors to estimate the parameters.

## 4. Results

### 4.1. The Temporal Trend in Los Angeles

A close examination of car theft data from Los Angeles suggests that what became known as the 'Kia Boys' vulnerability was discovered before the emergence of any recognizable social media trend. We focus on Hyundai Sonatas as representative of the class of vulnerable vehicles (see Table 1 and Appendix F). Fig. 3A shows weekly numbers of Hyundai Sonatas stolen in Los Angeles from Jan 1, 2017 to December 31, 2023 alongside Google Trends data on the term 'Kia Boys' in national Google searches and Los Angeles YouTube searches. Fig. 3B shows the spatial distribution of thefts in 2021 and 2023, before and after the 'Kia Boys' search spike. Through 2017-2019, Hyundai Sonata thefts were stable, averaging 37.6 thefts per week. A short period of suppressed thefts accompanied the start of COVID-19 lockdowns in March 2020, with 10 thefts per week, followed by a return to pre-lockdown theft volumes in July 2020, averaging 38.1 thefts per week. Theft recognizably surged in February 2021, rising to 94 thefts during the week of April 19, 2021. Theft in 2021 were broadly distributed across the city. Theft peaked during the summer of 2022, with a high of 134 cars stolen during the week of August 15, 2022, and settled around a new equilibrium for the remainder of 2022 and through 2023. Theft in 2023 rose across most neighborhoods with noticeable increased concentration in

South and East Los Angeles. The present study thus offers confirmatory local-level analysis of observations made with insurance claim data (Highway Loss Data Institute, 2024).

A logistic growth modeling (Tsoularis and Wallace, 2002; Bertrand, 2019) of the form  $N(t) = K_2(1 + e^{-R_0(t-c)})^{-1} + K_1$  fit to the weekly data charts a smooth transition between a pre-COVID-19 mean number of thefts of  $K_1 = 36.62$  thefts per week and post-‘Kia Boys’ mean of  $K_1 + K_2 = 91.17$  thefts per week, an increase in  $K_2 = 54.55$  thefts per week on average. Here  $R_0 = 0.06689$  is the basic reproductive number and may reflect the intrinsic rate at which knowledge of the vulnerability spread among Los Angeles car thieves. The parameter  $c = 243.5$  is the week number since the first week of 2017 (approximately August 26, 2021) during which the logistic growth curve starts to decelerate. The fastest phase of growth thus occurs more than a year before the first major spike in online interest in the ‘Kia Boys’ term nationally during the week of May 30, 2022 and in Los Angeles the week of Jun 6, 2022 (Fig. 3A). While content related to the ‘Kia Boys’ vulnerability spread rapidly online after June 2022 and impacted theft patterns in a number of cities, it appears to have played little role in the emergence of the theft problem in Los Angeles.

To support this last observation, we fit a vector autoregressive model (VAR) (Stock and Watson, 2001) to the data, focusing on the period after online searches for ‘Kia Boys’ first emerged in June 2021 (Fig. 3A). All three time series over this period are stationary (Dickey-Fuller and Phillips-Perron tests significant at  $p < 0.05$ ). Lag-order selection statistics suggest a model with three lags is appropriate. Pairwise comparisons suggest that lagged Hyundai Sonata thefts in Los Angeles predict (Granger-cause) U.S. Google searches for ‘Kia Boys’ ( $\chi^2 = 10.748$ ,  $df = 3$ ,  $p = 0.013$ ), but ‘Kia Boys’ searches do not predict Hyundai Sonata thefts ( $\chi^2 = 2.747$ ,  $df = 3$ ,  $p = 0.432$ ). There is no evidence for Granger-causality in either direction for Hyundai Sonata thefts and Los Angeles YouTube searches for ‘Kia Boys’ (thefts  $\rightarrow$  ‘Kia Boys’:  $\chi^2 = 5.974$ ,  $df = 3$ ,  $p = 0.113$ ; ‘Kia Boys’  $\rightarrow$  thefts:  $\chi^2 = 4.613$ ,  $df = 3$ ,  $p = 0.202$ ).

Pinpointing the initial discovery of the mechanical vulnerability by Los Angeles car thieves to the late summer of 2020, before the rise of the social media trend, is consistent with self-reported behavior in Milwaukee around the same time. In a widely-viewed YouTube profile of the ‘Kia Boys’ posted on May 31, 2022 Tommy G. (May 31, 2022), a Milwaukee youth states that he started stealing Hyundais and Kias “two summers ago,” which would date the start of the ‘Kia Boys’ phenomenon to the summer of 2020. We do not claim that the growth Hyundai Sonatas thefts in Los Angeles caused the viral ‘Kia Boys’ meme online nor do thefts in Los Angeles serve as a proxy for when this meme first appeared. Once established, however, the spread of the ‘Kia Boys’ meme could have reinforced Hyundai and Kia thefts in Los Angeles. The evidence for such an effect is mixed, however. On the one hand, the number of TikTok videos posted with the tag ‘Kia Boys’ increased over 2022

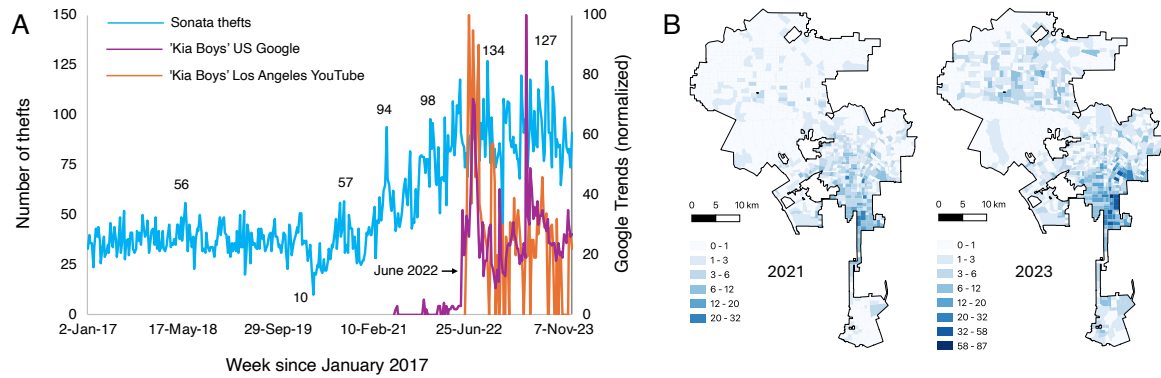
and 2023 as did the number of car thefts nationally (Fig. 1A) (Tapp and Coen, 2024; Highway Loss Data Institute, 2024). On the other, relative stability in the number of Hyundai Sonata thefts in Los Angeles through 2022-2023 suggests the saturation of any role played by social media.

## 4.2. Historical Analogs

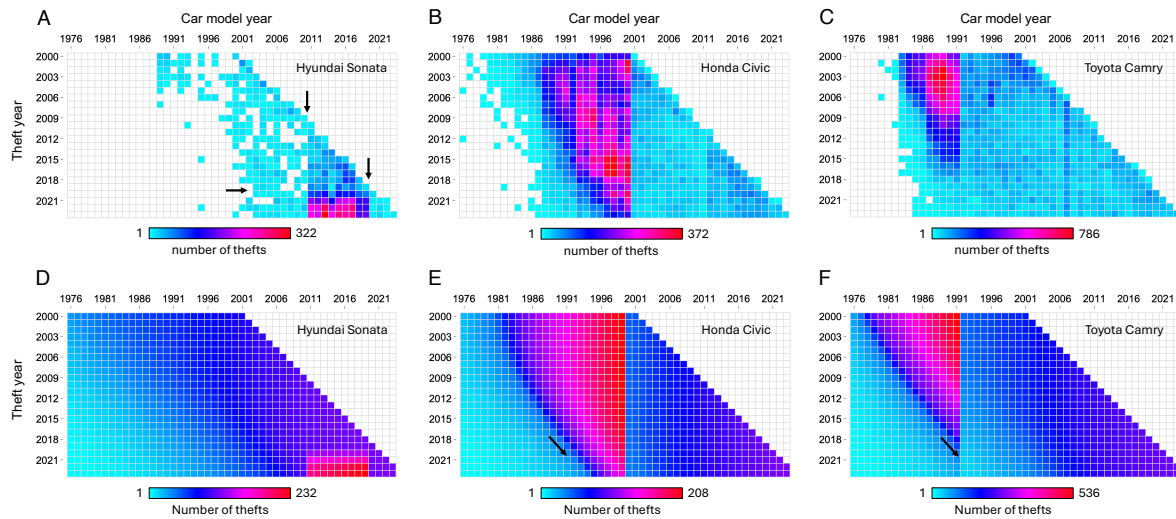
The long-term consequences of the knowledge of ‘Kia Boys’ vulnerability may be substantial regardless of how that knowledge was acquired by individual car thieves. Fig. 4A shows the immediate impact of the steps taken by Hyundai America to eliminate the vulnerability among new Hyundai Sonata model-years (Supplemental Movie S1). The matrix plot shows the frequency of thefts by car model-year and the year that the car was stolen (see Table 3 in Highway Loss Data Institute, 2024). The sharp boundary on either side of the vulnerable model-years (vertical arrows) is visually apparent as is the sharp rise in theft numbers beginning in 2020 (horizontal arrow).

The model-year theft pattern shown in Fig. 4A is insufficient to forecast the long-term impact of the ‘Kia Boys’ vulnerability. Assuming Los Angeles car thieves are now fully aware of the vulnerability, we turn to historical analogs to forecast long-term trends. Figs. 4B and C show temporal trends for thefts of Honda Civics and Toyota Camrys, respectively. These other well-known mid-sized sedans consistently ranked among the most frequently stolen vehicles in Los Angeles over the past 23 years, until Hyundai Sonatas surged to the top-ranked position (see Table 2). Both vehicles were known to be easy to steal with as little as a flat-head screwdriver (Light et al., 1993), a vulnerability that changed only after the manufacturers adopted chip-based ignition systems in 2001 and 1992, respectively. Pre-immobilizer Honda Civics provide a comparative benchmark for what vulnerable Hyundai Sonata thefts might look like 23 years after the ‘Kia Boys’ vulnerability was eliminated by the manufacturer, while pre-immobilizer Toyota Camrys provide a comparative benchmark after 32 years.

The security updates to Honda Civics and Toyota Camrys dramatically reduced the risk of theft for later model-years of each car type (Farrell, 2025; Brown, 2004). Vulnerable model-years without the update remained preferred targets for car thieves long after those cars were first sold, a pattern previously recognized by Brown and Thomas (2003) following the introduction of immobilizers in the U.K. For Honda Civics, this is seen in the sharp vertical discontinuity in the numbers of thefts of 2000 model-year Honda Civics, the last vulnerable year, and the 2001 model-year with new security features (Fig. 4B) (Supplemental Movie S2). The vulnerable 2000 model-year Honda Civic averaged 187.9 (sd 67.6) more thefts per year than the adjacent 2001 model-year Civics with immobilizers (max +325 in 2016; min +104 in 2023)(Fig. 4B). A similar discontinuity is observed for Toyota Camrys (Fig. 4C, Supplemental Movie S3). Lacking immobilizers, 1991 model-year Camrys were stolen in excess of the 1992 model-year, with immobilizers, every year between 2000 and 2018 (mean +202.6; sd 136.5; max +387



**Figure 3:** Trends in Hyundai Sonata thefts in Los Angeles and the viral spread of 'Kia Boys' content online. A. Weekly number of Hyundai Sonata thefts in Los Angeles and the normalized volume of searches for 'Kia Boys' in national Google Trends web and Los Angeles YouTube searches 2017-2023. B. Spatial distribution of Hyundai Sonata thefts across Los Angeles in 2021 and 2023.



**Figure 4:** Empirical and model-based matrix-frequency plots for cars stolen by model-year and year of theft. A-C. Thefts recorded by the LAPD between 2000-2023. D-F. Model-based estimates of thefts. A. and D. Hyundai Sonata. B. and E. Honda Civic. C. and F. Toyota Camry. Vertical arrows in A. mark the beginning and end of the window of vulnerable Hyundai Sonata model-years 2011-2020. Diagonal arrows in E and F indicate the direction of compression of the distribution due to vehicle scrapping.

in 2005; min +19 in 2018). In 2019, five more 1992 Camrys were stolen than 1991 Camrys for the first time in 20 years of observation. However, by this point—27 years after the introduction of chip-based immobilizers—very few cars of either model-year were being stolen at all; in 2019 there were 11 of the 1991 model-year Camrys reported stolen in Los Angeles and 16 of the 1992 model-year, against a total of 17,434 vehicles of all types stolen that year (see Table 2). Using traffic accidents as a proxy for environmental abundance, it is clear that very few of either the 1991 or 1992 model-year cars remain on the streets in 2023 (Bento et al., 2018) (Supplemental Movies S2 and S3).

Figs. 4D-F show the results of fitting the preferred mechanistic model introduced above in Equations (1) and (4) (Appendix C). For the simpler case involving Honda Civics and Toyota Camrys, the model captures the sharp discontinuity observed with the introduction of immobilizers (Figs. 4E and

F). It also captures the pressure of both market expansion and scrapping of older vehicles on theft patterns in any single year and across time. In the first row of Fig. 4E, for example, there is a gradual upward acceleration from very few thefts of older model-years (cyan) towards high numbers of thefts of younger model-years up to the discontinuity (red). The model expects that older model-year cars are rarely stolen both because fewer of those cars were sold new and because many of the cars that were sold initially were scrapped by the time of observation. Newer model-year cars are stolen more frequently under the guidance of the same mechanisms. The aggregate effect over time is the gradual concentration of thefts among a dwindling population of vulnerable vehicles (arrows in Figs. 4E and F), bounded on the right by the introduction of immobilizers and squeezed from the left by the pressure of scrapping. Finally, the model captures the point at which a theft hotspot collapses, recognized where thefts



of the last vulnerable model-year are no more numerous than thefts of the next model-year in line. For Toyota Camrys this is visible both in the empirical data and in the model fit around 2018 – 2019 (Figs. 4C and F). Supplemental Movies provide a dramatic illustration of the increasing compression of thefts around the youngest remaining vulnerable model-years (Movie S2) and the eventual collapse of thefts as those cars finally become unavailable in the environment (Movie S3).

The emerging dynamics for Hyundai Sonatas are conceptually similar to both the Civic and Camry cases. The one key difference is that the vulnerable cars are bounded both from above and below by immobilizer technologies that were removed for a certain period of time. Save for this one difference, we expect the dynamics of market expansion and scrapping to be the same going forward as observed for Honda Civics and Toyota Camrys. In the model we therefore observe a sharp concentration of thefts over model-years 2011 to 2019, with the slight acceleration over the range as a result of both market growth and scrapping of older cars (Fig. 4)(Supplemental Movie S1).

In Appendix E, we compare our baseline fitted models for Hyundai Sonata, Honda Civic, and Toyota Camry forecasts, determined by Equation 4, to a naive  $k$ -step ahead forecast using the mean absolute scaled error (MASE). For Hyundai Sonatas, the MASE score shows the mechanistic approach performing similarly to the naive approach over short a time horizon (1-year), which is understandable since the mechanistic model is not designed to capture short-term fluctuations in the intensity of thefts. The Sonata model performs substantially better over longer time horizons (5-10 years). The naive approach also does well in short-term forecasts of Civic and Camry thefts, with the mechanistic model again performing better over long time horizons (5-10 years).

### 4.3. Forecasting Future Impact

We use a combination of empirical comparisons and model projections to forecast the long-term impact of the ‘Kia Boys’ vulnerability in Los Angeles. The empirical observations are straightforward and can be gleaned from Fig. 4. Specifically, the concentration of thefts on vulnerable pre-2001 model-year Honda Civics had yet to dissipate by 2023, 23 years after chip-based immobilizers were introduced. In 2023, model-year 2000 Honda Civics were still being stolen at significantly higher rates than model-year 2001 Civics. For Toyota Camrys, by 2023 sufficient time has passed to observe the impact of chip-based immobilizers run its full course. The concentration of thefts on vulnerable pre-1992 Camrys was visible through at least 2018. In other words, model-year 1991 Toyota Camrys were still being stolen at higher rates than model-year 1992 Camrys in 2018, a full 27 years after the introduction of chip-based immobilizers.

Fig. 5A presents model-based forecasts for Hyundai Sonata thefts citywide by model-year from 2018 through 2050. The forecast suggests that thefts peaked in 2024, with a projected 1,783 total thefts of the vulnerable Sonata

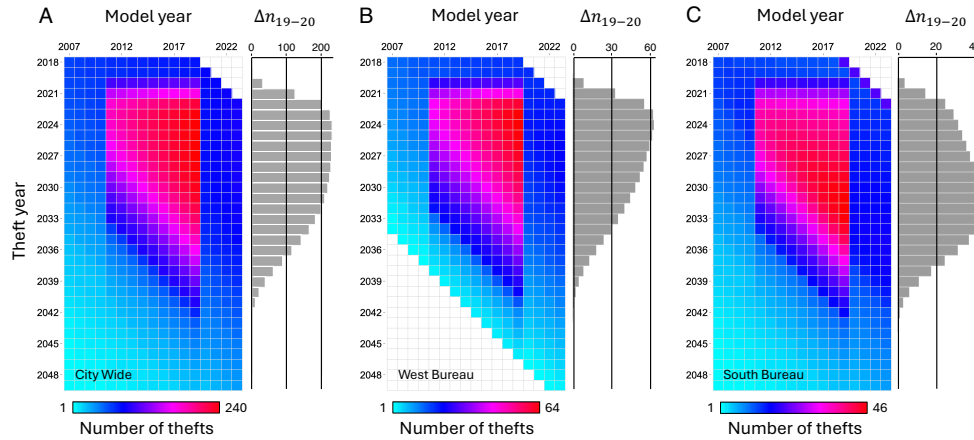
model-years (2011-2019) that year. We expect 235 more thefts of model-year 2019 cars than model-year 2020 cars in both 2024 and 2025. In 2032, thefts of the vulnerable cars citywide will fall below 1,000 units per year for the first time ( $n = 965$ , 203 more 2019 model-year thefts than 2020 model-year) and below 100 units only in 2039 ( $n = 82$ , 41 more 2019 model-year Sonata thefts than 2020 model-year). Projected thefts of the vulnerable model-years will finally fall below ten units citywide in 2042 ( $n = 7$ , five more thefts of 2019 model-year cars than 2020 model-year cars), 23 years after the the vulnerability was first learned by Los Angeles car thieves in 2020. Overall, the model projects 15,970 thefts of vulnerable model-year Hyundai Sonatas between 2024 and 2042.

In Appendix F, we extend the model to include impact of manufacturer recalls to fix the security vulnerability (Kia Hyundai Vehicle Theft Litigation, 2023a; Highway Loss Data Institute, 2024; Compton, 2025). Evidence suggests that the completion rate for recalls declines as a function of vehicle age relative to the date of the recall (NHTSA, 2021). We expect, therefore, that recalls issued by Hyundai and Kia in response to the 2022 class-action lawsuit to address the security vulnerability will have a much higher completion rate among 2019 model-year Hyundai Sonatas (four-years old in 2024) compared with 2011 model-year Sonatas (12-years old in 2024). For a recall that takes full effect in 2024, we expect thefts of vulnerable model-year Sonatas to have peaked in 2023. In 2024, the model suggest that theft volumes of Sonatas might be 40.7% lower than if no recall were issued (Appendix F, Fig. 10B). However, the projected impact of recalls is attenuated over time. With a 2024 recall, thefts of vulnerable model-year Sonatas are expected to fall below 1,000 units for the first time in 2026 ( $n = 997$ , 91 more 2019 model-year thefts than 2020 model-year), six years earlier than without a recall. Theft of vulnerable Sonatas are expected to fall below 100 units in 2038 ( $n = 66$ , 24 more 2019 model-year thefts than 2020 model-year thefts), and below 10 units in 2041 ( $n = 8$ , four more 2019 model-year thefts than 2020 model-year), in both cases just one year earlier than without a recall. The overall number of thefts with a 2024 recall is projected to be about 44% lower ( $n = 8,955$ ) between 2024 and 2042 with a recall. Appendix F also presents alternative scenarios where the recalls are alternatively issued in 2025 or 2026, more consistent with the multi-state settlement agreed upon in 2025. Delaying recalls increases the expected total number of thefts over time while also prolonging the period of excess thefts closer to conditions expected without a recall.

### 4.4. Impact on Communities

Our forecasting model suggests that the ‘Kia Boys’ vulnerability will continue to have an impact on Los Angeles car thefts through at least 2042. However, examination of the spatial trends in Honda Civic thefts between 2000-2023 suggests that Los Angeles communities will be differentially impacted over that time. Fig. 6 disaggregates the matrix plot for Honda Civics shown in Fig. 4B according to whether the





**Figure 5:** Projected Hyundai Sonata thefts by model-year from 2018-2049 and the difference in thefts  $\Delta n$  between the last vulnerable model-year 2019 and model-year 2020, when chip-based ignition immobilizers were reintroduced. A. Citywide forecast for all of Los Angeles. B. Forecast for LAPD's West Bureau. C. Forecast for LAPD's South Bureau.

thefts occurred in the LAPD's West or South Bureau. Theft of vulnerable Honda Civics—particularly the 2000 and 1999 model-year cars—was concentrated in West Bureau in the period 2001-2003 (Supplemental Movie S4). The hotspot then moved to South Bureau in 2015-2018. The peak in Honda Civic thefts in West Bureau targeted cars that were just one or two years-old. By contrast, the theft peak in South Bureau targeted cars that were already 15 years-old. This is evidence that car theft hotspots follow the use life-cycle of cars across both time and space. Specifically, new cars tend to be sold and driven in higher-income areas of a city. As cars age, they are resold multiple times and, through that market process, migrate to lower-income areas of a city. In 2022, West Bureau had a median household income of \$88,471. The median household income in South Bureau was a third lower at \$57,276. Thus, the impact of thefts of vulnerable model-year Honda Civics migrated over time from higher- to lower-income areas of the city and continues to linger in those low-income areas of Los Angeles. Hodgkinson, Andresen and Farrell (2016) observed a similar geographic shift in car theft hotspots from higher- to lower-income areas in Vancouver, British Columbia, between 2003 and 2013. Supplemental Movie S4 provides a vivid illustration of the shift in thefts from West to South Bureau beginning in 2013.

Figs. 5B and C disaggregate the citywide forecast of Hyundai Sonata thefts into expected patterns for LAPD's West and South Bureaus, respectively. In West Bureau, the model projects that thefts of all vulnerable model-year Sonatas peaked in 2023 at 445 total thefts. The model projects a peak difference of 63 more 2019 model-year cars stolen compared to the 2020 model-year in 2024 (Fig. 5B). Theft will fall below 100 units in West Bureau for the first time in 2034 to ( $n = 99.5$ , 30 more 2019 model-year Sonata thefts than 2020 model-year) and below 10 units in 2039 ( $n = 9$ , five more 2019 model-year Sonata thefts than 2020 model-year). The hotspot is expected to fully dissipate in West Bureau by 2042. In total, the model projects 3,213

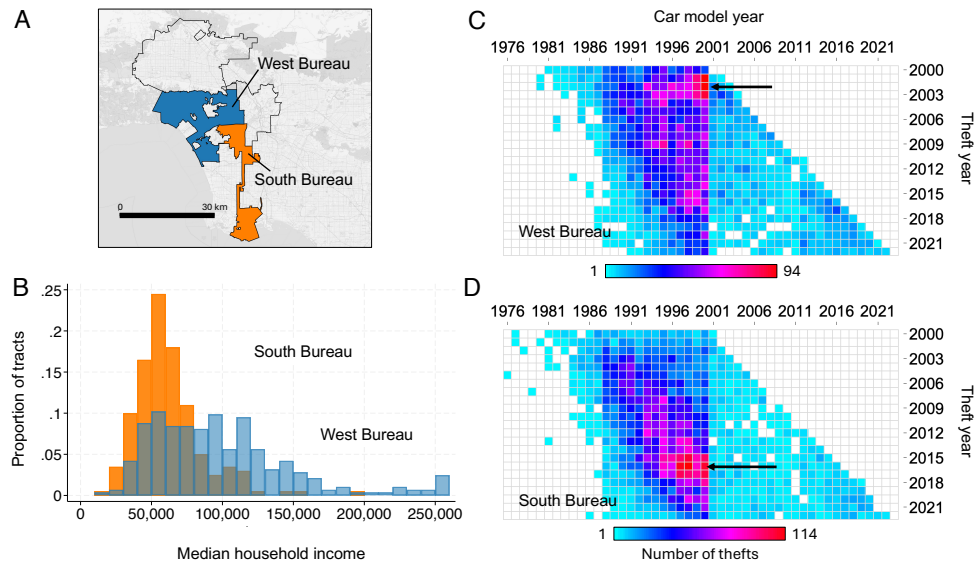
thefts of vulnerable model-year Hyundai Sonatas in West Bureau between 2024 and 2042.

In South Bureau, by contrast, thefts of vulnerable Sonatas are expected to peak in 2027 at 330 total vehicles (Fig. 5C). The peak difference in thefts between model-year 2019 and 2020 cars is not expected to occur until 5 to 6 years after the peak in total thefts. The model projects 46 more thefts of the last vulnerable model-year in 2032. Total thefts are expected to fall below 100 units for the first time in 2036 ( $n = 93$ , 31 more 2019 model-year Sonata thefts than 2020 model-year) and below 10 units in 2041 ( $n = 5$ , three more 2019 model-year Sonata thefts than 2020 model-year). The hotspot in South Bureau is expected to dissipate by 2043. In total, the model projects 3,459 thefts of vulnerable model-year Hyundai Sonatas in South Bureau between 2024 and 2042. Overall, approximately 246 more vulnerable model-year Sonatas are expected to be stolen in LAPD's South Bureau compared to West Bureau over the long-run.

## 5. Discussion

Our findings suggest that the narrative surrounding the emergence of the 'Kia Boys' vulnerability and its impact on thefts of many Hyundai and Kia car models is not as simple as widely portrayed. The story popularized by mainstream and social media during the summer of 2022 framed the problem as a viral online threat spilling over into offline settings. The implication is that, absent the online environment, the threat would not exist, or at least would not exist at the scale observed. This is an important counterfactual question central to understanding a wide range of offline harms (Williams, Tzani and Ioannou, 2023; Hartanto, Quek, Tng and Yong, 2021).

While it does appear that the 'Kia Boys' vulnerability was not widely known among the car thieves in some cities prior to widespread media reporting in 2022, in other cities



**Figure 6:** Median income distributions and Honda Civic theft patterns in LAPD West and South Bureaus. A. Location of LAPD's West and South Bureau. B. Median household income distribution in West and South Bureau by Census tract in 2023. C. Number of thefts of Honda Civics by model-year in West Bureau from 2000-2023. D. Number of thefts of Honda Civics by model-year in South Bureau from 2000-2023.

the vulnerability appears to have been discovered years before the viral spread of the 'Kia Boys' meme, without an obvious contribution of social media. In Los Angeles, the main focus of the present work, the discovery likely was made in mid-2020 through trial-and-error exploration by car thieves and then spread by word-of-mouth over the course of the next two years. We suspect but cannot prove that the social disruption provided by COVID-19 lockdowns facilitated the discovery of the vulnerability. Thus, the aggregate pattern across the U.S. therefore mixes some situations where social media may have played a decisive role and others where it likely did not. It is possible that the vulnerability was discovered even earlier in some places, given that immobilizers were absent beginning with the 2011 model-year for some Hyundai and Kia models. Close inspection of the car theft data for other cities not discussed here may reveal these very early discoveries. Whatever role social media played in spreading knowledge of the vulnerability more recently, it is likely the case that this role has now ended. The 'Kia Boys' vulnerability is now 'baked in' to offline theft dynamics.

Our models forecasting the long-term impact of the 'Kia Boys' vulnerability suggest that we will still be dealing with the consequences for years to come. In the baseline case, thefts of vulnerable Hyundai Sonatas are expected to have peaked in 2024 and but will remain elevated relative to non-vulnerable model-years through 2042. After 2042, the volume of thefts of the vulnerable model-years will be similar to non-vulnerable model-years. Similar forecasts pertain for vulnerable Kia Optimas, suggesting that the forecasting models are useful in describing risks across the full class of vulnerable vehicles (see Appendix D). Importantly, recalls issued by the manufacturers to address the 'Kia Boys' vulnerability may substantially reduce the total number of

thefts over the long-term. However, these recalls are not expected to reduce substantially how long the tail of elevated thefts will be. For example, uptake of security fixes at the rate seen historically for other recalls is expected to shorten the period of elevated thefts for Hyundai Sonatas by just one year, falling below 10 units in 2041 rather than 2042.

Our analysis also suggests that the impact of the 'Kia Boys' vulnerability will vary across communities in Los Angeles. Higher-income areas of cities are expected to move on from the spike in thefts as vulnerable model-years are replaced by new cars that lack the vulnerability. By contrast, lower-income areas may continue to see higher thefts for longer period as used cars with the vulnerability are sold into these communities.

The present study has several policy implications. First, the results add to the overwhelming evidence that chip-based engine immobilizers are an extremely effective car anti-theft technology (Farrell, 2025; Brown, 2015; Clarke and Harris, 1992). While there is some reason to worry that today's thieves will gain ready access to sophisticated tools to defeat the high-tech security of modern cars (Jacobson, 2023; Vardy, 2025), the 'Kia Boys' example suggests that most car theft remains an opportunistic and low skilled endeavor. Engine immobilizers present enough of a challenge for most thieves that they successfully deter (or at least thwart) most thefts. Yet, U.S. law still does not require immobilizers as standard equipment on new cars. It is possible that large tort settlements like those arising over the 'Kia Boys' vulnerability, may create incentives for other manufacturers to make immobilizers standard equipment voluntarily. However, it is disputed whether class action law suits stimulate voluntary corrective action by an industry at large (see Fitzpatrick,

2017). It is more likely that such suits generate public awareness that can indirectly spur new government regulation. The present example suggests that updates to the U.S. Motor Vehicle Theft Law Enforcement Act and Anti-Car Theft Act to require immobilizers as standard equipment would be a reasonable precaution against future emergent theft spikes like the 'Kia Boys' case.

Second, even with engine immobilizers adopted as standard equipment on all new cars, thefts of existing vulnerable cars may remain a challenge for years. The example of pre-1992 Toyota Camrys suggests that vulnerable cars can remain preferred targets for nearly three decades after they first hit the market. The related model forecasts for Hyundai Sonatas suggest elevated thefts through 2042. Security recalls may attenuate the period of risk, but not eliminate it. Following examples aimed at eliminating high-emission older vehicles to improve quality (Rodrigues, 2025), City- or state-level policies could offer incentives for scrapping older vulnerable vehicles. However, since the cost of theft is borne primarily by individuals and their insurers, it seems more plausible that the scrapping rate might rise naturally through increased costs of insuring vulnerable cars (Bennett and Wong, 2023).

Finally, crime prevention efforts need to deal with possible geographic shifts in demand over time. The research presented here suggests that car theft hotspots migrate geographically from higher to lower-income areas of a city (see Hodgkinson et al., 2016) as byproduct of vehicles following a natural life cycle from new purchase to used car and eventually to scrap. In Los Angeles, the balance of Hyundai Sonata thefts is expected to shift from West to South Bureau in 2028 and reach a peak difference in South Bureau around 2032. Policy makers thus have some time to prepare crime prevention and enforcement demands shifting to South Bureau. The response could be a straightforward as engaging the local communities to make car owners aware of the risks so that they can take appropriate action (Wortley and Denning, 1998). This might be particularly important as media coverage of the 'Kia Boys' vulnerability fades.

There are a number of limitations to this study that are worth noting. We have tried to account for the possibility that car owners will take advantage of security fixes offered by Hyundai and Kia. The class action law suit against Hyundai and Kia, settled in 2022, mandates software updates free of charge that rectify the vulnerability whenever authorized dealers come into contact with an eligible car (Kia Hyundai Vehicle Theft Litigation, 2023a). The multi-settlement announced in December 2025 mandates replacement of ignition cylinders in affected vehicles. We extended the baseline model to include recall completion rates based on data model-year age for a wide range of manufacturer recalls (NHTSA, 2021; Malec, Smith and Smuts, 2021). This model shows that, if recalls are eventually completed at the historical rate, the total number of thefts could be cut nearly in half, but time horizon for elevated thefts of vulnerable cars remains roughly the same as without a recall. However, it is possible that the completion rate for the

'Kia Boys' vulnerability will be higher than for other types of manufacturer recalls. Model forecasts will need to be adjusted as more information becomes available.

We have not controlled for macroscopic trends in car theft rates unrelated to specific vulnerabilities in our forecasts. Long-term theft rates are impacted by several factors including changes in offender populations and the role that cars play in routine, day-to-day activities (Rosenfeld, 2024; Copes, 1999; Mohler, Bertozzi, Carter, Short, Sledge, Tita, Uchida and Brantingham, 2020). The forecasts present here are therefore subject to change given shifts in these macroscopic conditions. However, since our forecasts concern thefts only of the vulnerable Hyundai Sonata model-years, produced between 2011-2019, the environmental abundance and vulnerabilities of more recent model-years (including those yet to be produced) are irrelevant. Nevertheless, future mechanistic models might seek to include demographic and routine activity features to better anticipate general changes in crime environments (Rosenfeld, 2024).

## 6. Appendix A: Data, Materials, and Software Availability.

Aggregated data by car model-year and theft year along with R and STAN are deposited in GitHub (<https://>). All other data and additional analyses are included in the manuscript and/or SI Appendix.

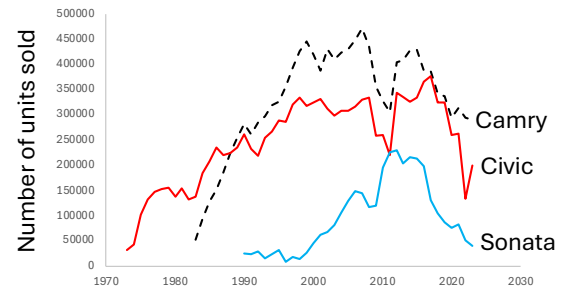
### 6.1. Appendix B: Traffic Collisions as a Proxy for Abundance

The environmental abundance of specific make-models of cars plays an important role in long-term theft dynamics. Here we suggest that traffic accidents provide a suitable proxy for the relative availability of different car make-model-years in the environment. Specifically, the number of traffic accidents reported to the police by make-model-year of car is a function of the number of such cars present in the environment, their road activity, and the reporting behavior of parties involved in the accident. We assume that factors such as driver skill are randomly distributed within make-model-years. The number of cars in operation is determined by the number joining the fleet through sales of new cars and the number leaving through scrapping, which has an impact that compounds as cars age (Bento et al., 2018). The road activity of car is also a function of car age, with newer cars systematically seeing more miles than older cars (Dutt, Reinfurt and Stutts, 1977; Cooper, Osborn and Meckle, 2010). Finally, we expect that the willingness to report a traffic accident to the police skews towards newer model-years, consistent with the observation that the costs of calling the police are weighed against the cost of the crime or accident (Bowles, Garcia Reyes and Garoupa, 2009; Brantingham, Mohler and Vorobeychik, 2025). We expect vehicles to be stolen in proportion to their availability conditional on any unique vulnerabilities that might make some make-model-years to be more at risk than others.

Supplemental Movies S1, S2 and S3, show the number of reported thefts and accidents (y-axis) by make-model-year (x-axis) for Hyundai Sonatas, Honda Civics, and Toyota Camrys, respectively. Thefts of and accidents involving Hyundai Sonatas in Los Angeles were extremely rare until the 2011 model-year when accidents begin to rise (Movie S1). This rise may reflect an increase in the number of Hyundai Sonatas sold new as indicated by national sales figures (see Fig. 7). Thefts of Hyundai Sonatas remain subdued until 2020 when the shape of the distribution first resembles the distribution of accidents. Thefts dramatically exceed accidents in each year from 2021-2023, suggesting a substantial shift in thieves targeting a largely stable abundance of vehicles in the environment. The concentration of thefts in 2020-2023 is clearly restricted to the vulnerable 2011-2019 model-years.

Honda Civics provides a clear demonstration of the impact of both the introduction of immobilizers and scrapping of older cars (Supplemental Movie S2). At the outset of the observational time window in 2000 the accident and theft distributions are similarly shaped, with a sharp leading edge positioned at the concurrent model-year, which includes a handful of 2001 cars introduced at the end of 2000. The theft and accident distributions both have tails that decay towards older model-year cars. We conclude that the distribution of thefts largely follows the same abundance constraints as accidents. Beginning with the 2001 observation year, we see a divergence between the accident and theft distribution. The leading edge of the accident distribution continues to move forward in time, retaining a characteristic skewed shape, while the theft distribution develops an sharp boundary that is stationary at model-year 2000. By 2016, 53.7% of the mass of the accident distribution was concentrated among post-2000 model-years, while 98.2% of the theft distribution was concentrated among pre-2001 model-years lacking an immobilizer. A small leading edge of thefts post-2000 model-year Civics moves forward in time reflecting limited numbers of thefts of post-2000 Civics that are much harder to steal. The theft distribution begins to dissipate after 2017 as model-year 2000 and earlier Honda Civics become harder to find, a situation documented by the substantial reduction in accidents involving these early model-year cars (Movie S2).

The animation for Toyota Camrys provides a guide of what may lie in store for Honda Civics and Hyundai Sonatas (Movie S3). Toyota introduced engine immobilizers for the Camry in 1992, meaning that this theft-prevention device had already been in place for nine years by the first year of observational data in 2000. The accident distribution had already diverged substantially from the theft distribution in 2000 and thefts were strongly concentrated among the youngest of the vulnerable model-years (i.e., model-years 1988-1991). The concentration among vulnerable model-years peaked in 2003—11 years after the introduction of immobilizers. Thefts then began to collapse such that the special status of the pre-1992 Toyota Camry among Los Angeles car thieves is largely gone by 2018. The distribution



**Figure 7:** Number of car units sold in the United States per year since entry into the market for Hyundai Sonatas, Honda Civics and Toyota Camrys. Data assembled from marketing materials published by each of the manufacturers.

**Table 3**

Model comparison for different specifications of  $h_m$  fit to Sonata data.

| Model    | Log Likelihood | # Parameters | AIC       |
|----------|----------------|--------------|-----------|
| power    | 22076.74       | 8            | -44137.48 |
| exp      | 22030.40       | 8            | -44044.80 |
| constant | 21954.69       | 7            | -43895.38 |
| sales    | 22215.63       | 9            | -44413.25 |

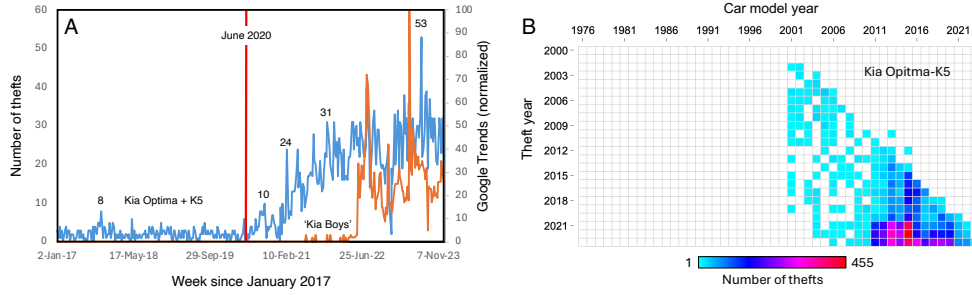
of accidents suggests that the lack of availability of cars was a primary driver for the reduction in thefts of vulnerable Camrys.

## 7. Appendix C: Alternative Model Forms

Here we compare alternative specifications for the sales trend function  $h_m$  included as a term in Equation (1). We compare power  $h_m = (t_m - 1975)^d$ , exponential  $h_m = \exp(d(t_m - 1975))$ , constant  $h_m = 1$  and Gaussian functions  $h_m = \exp(-(s_m - \mu)^2 / (2\sigma^2))$ , where  $t_m$  is the year for each model-year and  $h_m$  is the expected number cars of that model-year sold. The Gaussian function uses observed sales data  $s_m$  in place of model-year  $t_m$  (Fig. 7).

Table 3 shows the relative performance of the alternative specifications of  $h_m$  when fitting Equations (1) and (4) to Hyundai Sonata thefts. The Gaussian function provides a better relative fit compared to other abstract functional forms for the Sonata theft data. The power model ranks second. Nevertheless, we favor the power model for three reasons. First, the model comparison statistics suggest that the power model is a good generic choice if car sales data are not available. From experience, car sales data are difficult to obtain at the local level. Second, Fig. 4 shows that the qualitative fit between the stylized power model and the historical data is good. In fitting the main model, we suspect that the effects of scrapping, captured by the function  $g_m$ , mostly override the subtle differences in alternative forms of  $h_m$ . Finally, since future sales are unknown, the stylized power model is a reasonable choice for how companies expect market share to grow over time, facilitating the projection of the model through 2050.





**Figure 8:** Trends in thefts of Kia Optima. A. Theft of Kia Optima per week during 2017-2023 plotted alongside Google Trends data for the search term 'Kia Boys'. Compare with Fig. 1A in the main text. B. Theft of Kia Optima by model-year recorded by the LAPD. Note: The Kia Optima was re-branded as the Kia K5 in 2021. Theft counts in A and B combine the Optima and K5 for the 2021-2023 years.

## 8. Appendix D: Replicating Results with the Kia Optima

In the main text, we focus on Hyundai Sonatas because they were the most frequently targeted vehicle in the class of vulnerable Hyundai and Kia models listed in Table 1. Here we show that our results generalize to other make-models in the class.

Like the Hyundai Sonata, the Kia Optima surged in popularity among Los Angeles car thieves beginning in 2020. The Kia Optima jumped to the 13th most frequently stolen car in that year and then rose to the number five and number three positions in 2021 and 2022, respectively. It occupied the number two position in 2023, immediately behind Hyundai Sonatas. It was not in the top 100 most stolen cars (rank 101) as recently as 2013. Fig. 8A shows that the increase in thefts followed a very similar trajectory to Hyundai Sonatas, rising from a stable mean number of thefts per week (2.1), prior to June 2020, to a new weekly mean (26.4) after June 2022. The rise in thefts predates the emergence of the 'Kia Boys' meme. Fig. 8B shows that the theft hotspot is concentrated among the vulnerable model-years 2011-2020, falling sharply with the 2021 model-year when fixes were introduced by Kia.

Fig. 9 shows the expected number of thefts of Kia Optima through 2049 using the baseline model introduced in the main text. Theft of the vulnerable model-year cars are expected to have peaked in 2023 ( $n = 1,495$ ). Theft is expected to fall below 1,000 units for the first time in 2029 ( $n = 894$ , 181 more 2020 model-year than 2021 model-year cars), below 100 units for the first time in 2036 ( $n = 75$ , 39 more 2020 model-year than 2021 model-year cars), and below 10 units in 2039 ( $n = 4$ , three more 2020 model-year than 2021 model-year cars). The Kia Optima hotspot is expected to dissipate in 2041 one year earlier than the Hyundai Sonata hotspot under the baseline model.

## 9. Appendix E: Mean Absolute Scaled Error

Here we compare mechanistic forecasts based on Equation 4 in the main text with naive  $k$ -step ahead forecasts using the Mean Absolute Scaled Error (MASE) (Hyndman and

**Table 4**

MASE values for Sonata, Civic, Camry, and Optima forecasts and different year lags for the naive  $k$ -step ahead method.

| lag $k$ | Sonata | Civic | Camry | Optima |
|---------|--------|-------|-------|--------|
| 1       | 1.03   | 2.22  | 1.97  | 1.67   |
| 2       | 0.64   | 1.59  | 1.18  | 1.21   |
| 3       | 0.51   | 1.26  | 0.83  | 0.96   |
| 4       | 0.48   | 1.09  | 0.63  | 0.93   |
| 5       | 0.47   | 0.94  | 0.49  | 0.91   |
| 6       | 0.46   | 0.84  | 0.39  | 0.89   |
| 7       | 0.45   | 0.79  | 0.32  | 0.86   |
| 8       | 0.44   | 0.73  | 0.27  | 0.84   |
| 9       | 0.43   | 0.72  | 0.24  | 0.83   |
| 10      | 0.42   | 0.67  | 0.21  | 0.82   |

Koehler, 2006). Letting  $e_t^m = y_t^m - \hat{y}_t^m$  be the model forecast error in year  $t$  for model-year  $m$ ,  $y_t^m$  be the observed number of thefts in year  $t$  for model-year  $m$ , and

$$q_t = \frac{e_t^m}{\frac{1}{n-k} \sum_{i=k+1}^n |y_i^m - y_{i-k}^m|},$$

then the MASE is computed as  $\text{mean}(|q_t|)$ .

Table 4 displays the MASE values for Sonata, Civic, Camry, and Optima forecasts (see Figure 4). For Sonata, the model performs similarly to the naive method for short-term (1-year-ahead) forecasts. This is expected since our mechanistic approach is not designed to capture short-term trends in the data. Our mechanistic Sonata model performs significantly better than the naive approach for longer-term forecasts. For Civic, Camry, and Optima forecasts, the 1-step and 2-step ahead forecasts are better able to capture short-term spikes in thefts, however the mechanistic model outperforms the naive  $k$ -step ahead forecast for longer-term (5-10) year horizons.

## 10. Appendix F: Forecasting the Impact of a Recall

We consider an alternative model for Sonata thefts that takes into account a recall to fix the vulnerability. For this

purpose we use the recall completion rate model detailed in Fig. 10 of the National Highway Traffic Safety Administration (NHTSA) 2021 report to Congress on vehicle safety recall completion rates (NHTSA, 2021). The completion rate is modeled as a logistic function of the age of the vehicle at the time of the recall (i.e., older cars are less likely to be brought to a dealer for the work to be done). The completion rate  $w_r^m(t)$  in year  $t$  for model-year  $m$  and recall year  $t_r$  for Hyundais is given in Equation 8, with parameters  $C_1 = 100.511$ ,  $r_1 = -0.223$ ,  $x_1 = 7.968$ , and  $C_0 = -0.266$ . We plot the completion rate in Fig. 10A.

$$w_r^m(t) = C_0 + (C_1 - C_0)\sigma(r_1(t_r - t_m - x_1)) \quad (8)$$

$$f_m(t) = f_0 + \quad (9)$$

$$f_1 \chi\{t_m \geq s_1\} \chi\{t_m \leq s_2\} \chi\{t \geq u\} \cdot \sigma(r_0(t - u - x_0)) \cdot w_r^m(t) \quad (10)$$

We assume that the vulnerability is fixed either within the same year of the recall announcement or not at all. The intensity of Sonata thefts of model-year  $t_m$  in theft year  $t$  for a recall in year  $t_r$  is then reduced by  $w_r^m(t)$  (Equation 11).

In Fig. 10B, we plot the total forecasted Sonata thefts per year and compare different scenarios where the recall is issued in 2024, 2025, or 2026. For the 2024 recall, we project that Sonata would be cut in half overall but remain statistically elevated through 2041, only one year shorter than without a recall. Further outcomes of the recall model are described in the main text.

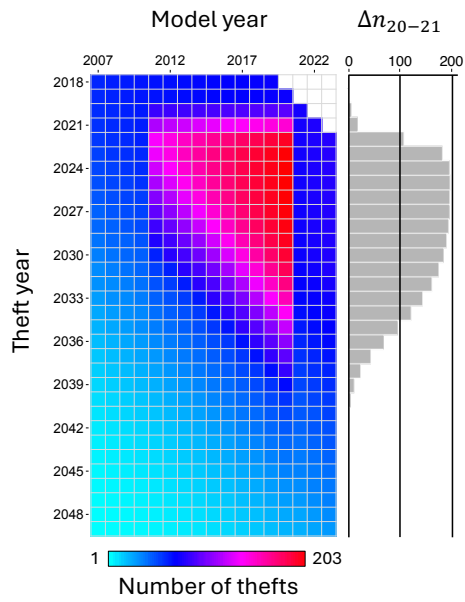
## CRedit authorship contribution statement

**P. Jeffrey Brantingham:** Conceptualization of this study, Methodology, Formal analysis, Writing - Original Draft, Review & Editing. **Craig D. Uchida:** Conceptualization of this study, Methodology, Formal analysis, Writing - Original Draft, Review & Editing. **Fred Morstatter:** Conceptualization of this study, Methodology, Formal analysis, Writing - Original Draft, Review & Editing. **George Mohler:** Conceptualization of this study, Methodology, Formal analysis, Writing - Original Draft, Review & Editing.

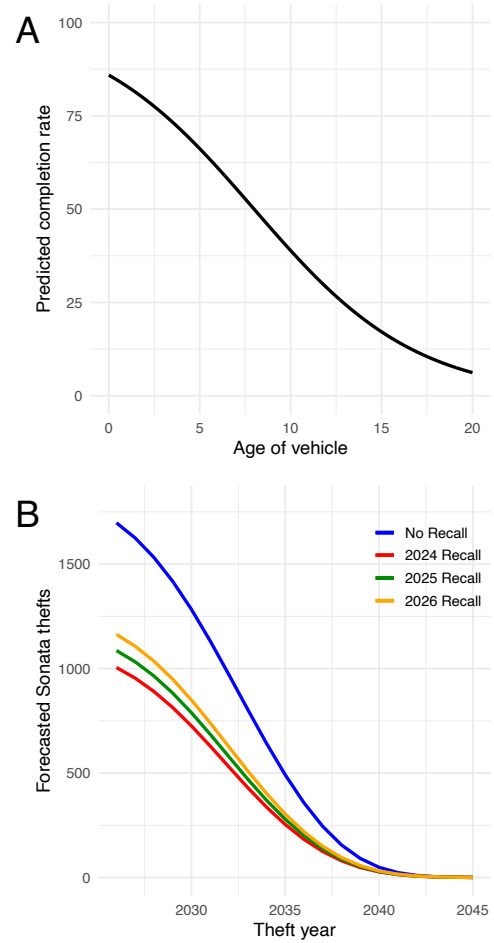
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**Figure 9:** Matrix plot of the number of Kia Optima thefts by model-year and year of theft. Theft counts combine the Optima and K5 for the 2021-2023 years.



**Figure 10:** Recall completion rates and their impact on thefts. A. The recall completion rate for all Hyundai make-models as a function of age of the vehicle at the time of the recall. B. Total forecasted Hyundai Sonata thefts per year for security recalls issued in different years.